

Full matter carrier

$$\mathcal{M} \simeq \mathbb{C}^{252}$$

$$\Lambda^6 \mathbb{C}^7 \oplus \Lambda^2 \mathbb{C}^7 \oplus \Lambda^4 \mathbb{C}^7$$

$7 + 21 + 35 = 63$; four Dirac components

$$\psi = 2Ev$$

χ : independent linear dual

$$A \in \text{End}_{\mathbb{C}}(\mathcal{M})$$

arbitrary A ; designated field and dual

Occupation response

$$\mathcal{E} = \mathbb{C}^8$$

$$v = \tfrac{1}{2}P\psi, \quad \langle v, v \rangle = 1$$

$$R(A) = \textcolor{brown}{S}PAE$$

$$PE = 1, \quad S^* = S, \quad S^2 = 1$$

E embed P project S swap

$R(A)$ need be neither unitary nor self-adjoint

Fermionic Fock space

$$\mathcal{F}_-(\mathcal{E}), \quad \dim = 256$$

$$I(v) \in \Lambda^1 \mathcal{E}$$

$\Lambda^0 \mathcal{E}$ 1	$\Lambda^1 \mathcal{E}$ 8	$\Lambda^2 \mathcal{E}$ 28
$\Lambda^3 \mathcal{E}$ 56	$\Lambda^4 \mathcal{E}$ 70	$\Lambda^5 \mathcal{E}$ 56
$\Lambda^6 \mathcal{E}$ 28	$\Lambda^7 \mathcal{E}$ 8	$\Lambda^8 \mathcal{E}$ 1

$$\chi(A\psi) = 4k \langle v, R(A)v \rangle = 4k \langle I(v), \text{d}\Gamma(R(A))I(v) \rangle, \quad k = \sqrt{2}$$

